

Tilak Madichetti

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Summary

Software engineer with deep Solidity and EVM security expertise. Built and led **Aderyn**, an open-source Rust static analyzer listed in the **official Solidity documentation** and supported by an **Ethereum Foundation grant**. Its 65+ vulnerability classes had to be defined precisely enough to execute as automated checks, then refined against findings from independent security researchers.

Self-taught, and now moving from analyzing smart contracts to authoring them — with **interest across the whole lifecycle, from design and testing through deployment, upgrade paths, and incident response**. Equally curious about the infrastructure around contracts: learned ReScript in 2025 in order to read through the source of Envio, a blockchain indexer.

Experience

Cyfrin — Software Engineer → AI Security Engineer

Remote · Jul 2024 - Jul 2026 · US-based web3 security startup, team distributed across the US, UK, Italy, Canada, Turkey, and India

Cygent — continuous AI security platform for smart contract protocols (*Jan - Jul 2026*)

- Built agent skills and agentic infrastructure for a platform that reviews pull requests, surfaces exploit paths, validates fixes at release, and triages bug bounty submissions.
- Developed AST-level analysis tooling for the platform; when the audit manager at Cyfrin built a skill-based agent harness that outperformed it, shifted to collaborating on that approach instead — extending coverage beyond Solidity to Rust and TypeScript.
- Delivered multi-tenant platform infrastructure with per-instance compute, service, and data isolation, backed by PostgreSQL — including container lifecycle management and orchestration for isolated tenant environments (Python, TypeScript, Docker).
- Full-stack development across the platform — SvelteKit frontend, TypeScript backend services, and PostgreSQL.
- Supported the in-house audit team, translating auditor workflows into platform capabilities.

Aderyn — open-source Solidity static analyzer (*external contributor since Dec 2023; full-time maintainer Jul 2024 - May 2025*)

Built under direct mentorship from **Alex Roan**, former smart contract lead at Chainlink Labs.

- Spearheaded roadmap and development of a Rust static analyzer with **65+ detectors** covering access control, reentrancy, and other high- and low-severity vulnerability classes, with SARIF, Markdown, and JSON output. **790+ GitHub stars, 189,000+ downloads**.
- **Listed in the official Solidity documentation, Foundry's docs, and ethereum.org** as a recommended Solidity static analyzer. Adopted by Audit Wizard to power automated vulnerability detection; cited by Nethermind's security team; architecture reviewed in BlockThreat.
- **Awarded an Ethereum Foundation grant** in recognition of the tool's contribution to Ethereum security tooling.
- Forked Foundry to build Abstract Syntax Tree extraction across multiple Solidity compiler versions; wrote native Rust bindings for compiler AST artifacts.
- Designed control flow graphs, call graphs, and a high-level IR over those ASTs, optimized for fast whole-codebase traversal so detectors run in bounded time and space.
- Partnered with independent security researchers and competitive audit teams to refine detector precision against findings from live protocols.
- Embedded an LSP server for real-time diagnostics on file save; shipped a zero-dependency VS Code extension in TypeScript (**2,500+ installs**) to remove runtime-install friction for Solidity developers.
- Built CI/CD publishing the Rust binary to npm and Homebrew via a custom cyfrinup installer for safe, versioned distribution of Cyfrin's security tooling.
- Embedded an MCP server to let AI agents traverse the high-level IR, giving in-house researchers deterministic code-path discovery instead of manual search.
- Mentored open-source contributors and built abstractions that shortened their test-and-submit loop.

Updraft & Cyfrin Profiles — web3 education platform, 1M monthly visits / 50k+ users (*Jun 2025 - Dec 2025*)

Full-stack development across a SvelteKit frontend with Tailwind CSS, TypeScript backend services, Prisma ORM, Redis caching, and PlanetScale.

- Implemented a TF-IDF search indexer in Rust with Axum for a performance-critical search surface.
- Built the Profiles Achievements system — streak badges, course completion badges, and certifications earned across exams, quizzes, and lessons — including a retroactive production database backfill to award existing users.
- Owned platform infrastructure and the CMS supporting timed course launches.
- Coordinated launches across six timezones with design, product, marketing, and external educators — scheduling releases to marketing-optimal windows and

gathering course material on deadline.

Siemens — Research Engineer

Bangalore, onsite · Jun 2023 – Jul 2024

- Rewrote a legacy Python data-transformation library in Rust, cutting runtime from **6+ hours to ~25 minutes** and unblocking faster staging iterations across multiple teams.
 - Led ideation and development of **GraphVerse** (Java, Spring Boot) — effectively version control for graph databases, and the first internal solution to change management of knowledge graph data. **Earned an internal grant from Siemens' Intellectual Property department.**
 - Built an end-to-end Elo-rating evaluation system for ranking in-house large language models.
 - Prepared knowledge graphs and training datasets for internal LLMs; onboarded and coordinated interns.
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Smart Contract & Security Work

Competitive audit findings (CodeHawks) — Identified **high and medium severity** issues in live DeFi protocols including QuantAMM, The Standard, and Swan Dria — six findings accepted by independent judges across three contests. Reasoning about access control, economic failure modes, and edge-case behaviour in contracts holding real value.

Solar (Paradigm) — 10+ merged pull requests to the next-generation Rust Solidity compiler, including a substantial portion of the **Abstract Syntax Tree validation logic**, work that directly affects the compiler's correctness and security checks.

Detector design at scale — Encoded 65+ distinct vulnerability classes into executable analysis, requiring precise operational definitions of reentrancy, access control failures, and unsafe upgrade and delegation patterns rather than heuristic familiarity with them.

Fingerprint — On-chain messaging application (Solidity, Alloy, Tauri, SvelteKit) using Diffie-Hellman key exchange for end-to-end encryption. Hackathon project; won Cyfrin's internal hackathon.

Technical Skills

Smart contracts & EVM — Solidity, OpenZeppelin (access control, upgradeable proxies, ERC standards), Chainlink, Foundry (forked and extended), Alloy, fuzzing with Foundry, Solidity compiler internals, AST/CFG/IR analysis, vulnerability classes and security patterns, SARIF tooling

Languages — Rust (async/tokio, axum, hyper, tower, sync primitives, cargo-dist and the wider tooling ecosystem), TypeScript (Node, SvelteKit, Tailwind CSS, Fastify, Prisma), Python, Go, Java (Spring Boot)

AI-assisted engineering — agent skills and harness design, MCP servers exposing structured codebase access to agents, provisioning isolated cloud environments for

autonomous agent execution; hands-on experience of where agent tooling outperforms deterministic analysis and where it doesn't

Platform & infrastructure — Docker, container lifecycle and orchestration, multi-tenant architecture, PostgreSQL, CI/CD, PlanetScale, Redis, Vercel; self-hosted bare-metal environment running DNS, databases, SSO (Authentik, Keycloak), certificate signing, hardware-backed key custody (Trezor), and a full observability stack (Grafana, Prometheus, Loki, Tempo, Alloy)

Currently exploring — Talos Kubernetes on AWS, Terraform, Cilium CNI, Envoy, Helm/Kustomize, Elixir/BEAM

Earlier Experience

Summer Research Intern, Siemens — *Remote, Bangalore* · May 2022 - Jul 2022 Automated extraction pipelines from JSON/XML/XLSX into Pandas, transformed to TTL/Cypher, loaded into Neo4J and GraphDB. Built dashboards for visual debugging of RDF data. Converted to full-time.

Full Stack Developer Intern, Edilitics — *Remote, Mumbai* · Oct 2021 - Dec 2021 Built a full-load pipeline from PostgreSQL to BigQuery using Airflow DAGs with Flask control endpoints. Proposed and executed a migration to Prefect, rewriting the ETL logic and building a React interface for user-composed pipelines.

Community

10 years on Stack Overflow — technical answers reaching 5.6M+ developers.